1. For the following function, design a minimum 2-level NOR-NOR network.

   $F(A, B, C, D) = \Pi M(1, 3, 7, 8, 9, 12, 14) + \Sigma d(0, 10, 13, 15)$

   Step 1: Draw the K-map

   Step 2: Find the minimum POS or SOP (circle one)

   Step 3: Draw the digital circuit

2. Given the function below, draw a digital circuit for an implementation which uses only four 2-input NAND gates.

$$G(A, B, C, D) = (A + B)(A + D')(B + C' + D)(A + D' + C)$$

Step 1: Simplify to remove one term.

Step 2: Draw the K-map and find the minimum POS or SOP (circle one)

Step 3: Factor the expression so every gate only has two inputs

Step 4: Draw the digital circuit